

Anemia and Bleeding Disorders with Dental Modifications

OMFS 512: Integrated Biomedical and Clinical Sciences

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

1. Applying the basic science knowledge to clinical application.
2. Recognize the physical and oral manifestations of the hematologic diseases
3. Establish a connection between the disease process and the oral manifestations
4. Using critical thinking skills to determine which patients are safe to treat in a dental setting.

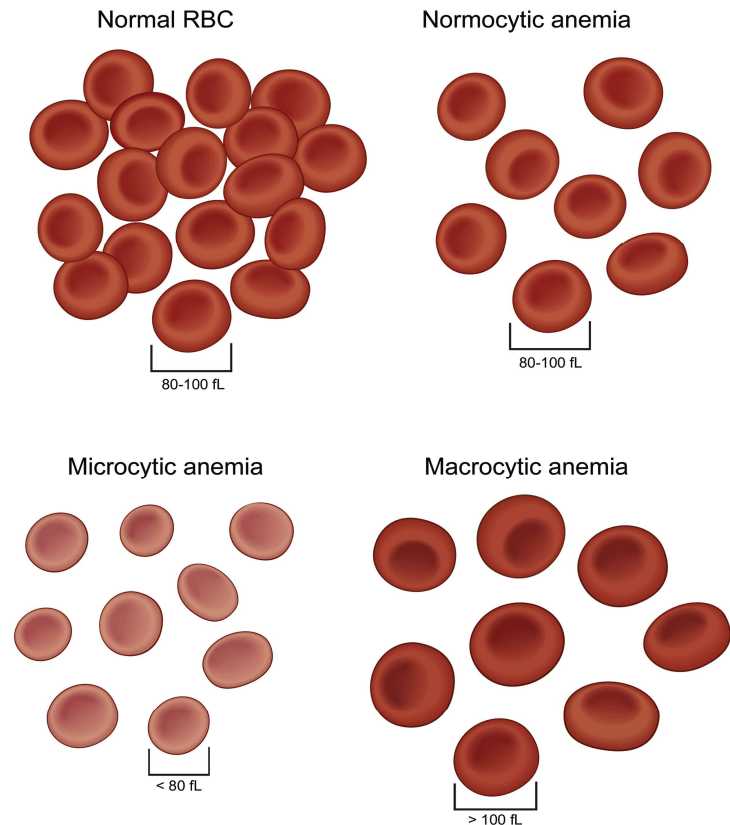
Competency statements met

1.3, 1.5, and 2.1

Types of Anemia

* Table 22.1 in textbook

- **Microcytic**
 - Iron-Deficiency Anemia
 - Thalassemia
- **Normocytic**
 - Sickle Cell Anemia
 - Glucose 6-Phosphatase Dehydrogenase
 - Aplastic Anemia
- **Macrocytic**
 - Folate (B9)-Deficiency Anemia
 - Vitamin B12 Deficiency & Pernicious Anemia
- **Anemia associate with Systemic Diseases**
 - Renal Failure
 - Hypothyroidism
 - Lead poisoning



Iron-Deficiency Anemia

- Most common form of anemia
- Female predilection
- Causes
 - Issues with RBC (increased destruction or decreased production)
 - Issue with low iron intake or absorption in the body
 - Excessive loss of blood
 - Increased demand for iron

Iron Deficiency Anemia

- Physical Manifestations: Pale skin, fatigue, and dyspnea, koilonychia(spoon-shaped nails), hair loss
- Oral Manifestations
 - Pallor, erythema of the oral mucosa, glossitis, glossodynia, aphthous-like ulcers, and angular cheilitis
- Health Concerns
 - Precancerous condition
 - Blood Transfusion

Iron Deficiency Anemia



Angular Cheilitis



Patchy glossitis of the tongue



Diffuse Glossitis "Beefy red" Tongue

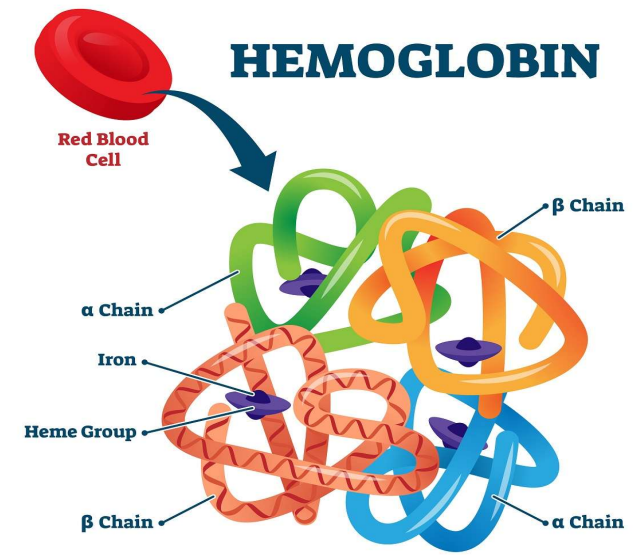


Erythema of the labial mucosa

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Thalassemia

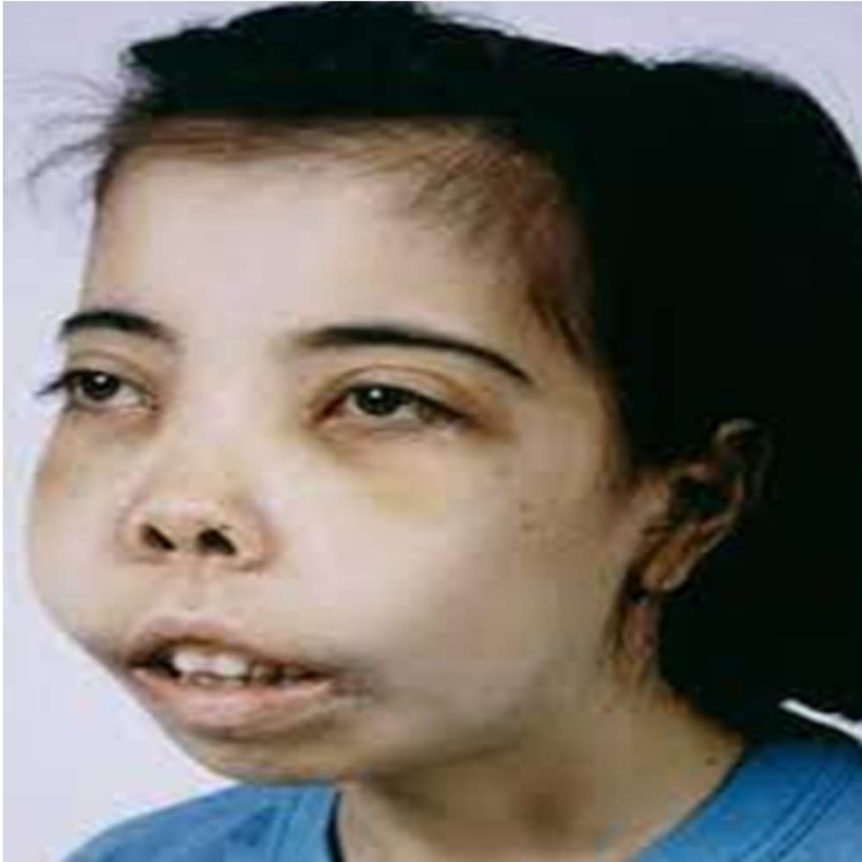
- Hemolytic anemia associated with patients of Middle Eastern, Mediterranean, and South Asian descent
- **Forms**
 - **Alpha-Thalassemia**
 - 4 genes total (2 genes from each parent)
 - Silent carrier, alpha-thalassemia trait, HbH disease, and Hydrops fetalis
 - **Beta-Thalassemia**
 - 2 genes (1 gene from each parent)
 - Beta-thalassemia minor, beta-thalassemia intermedia, **beta-thalassemia Major (Cooley's Anemia)**



Beta-Thalassemia Major

- Physical Manifestations
 - Jaundiced appearance, pallor, splenomegaly, and small stature
 - Bony changes “**hair on end**” appearance in skull radiographs
- Oral Manifestations
 - Bony changes in the jaws
 - **Bilateral maxillary protrusion with spacing of the teeth**
 - Swelling of the salivary glands and xerostomia

Beta-Thalassemia Major



Facial features of Beta-Thalassemia Major

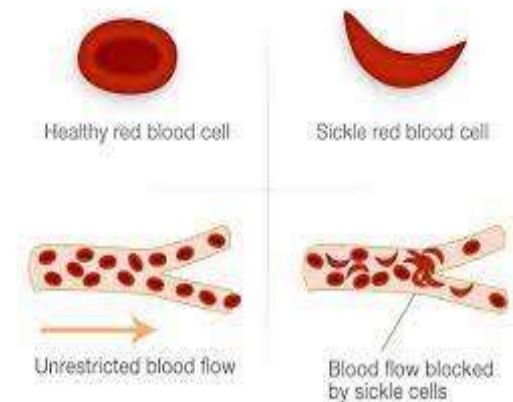


"Hair on end" radiographic appearance

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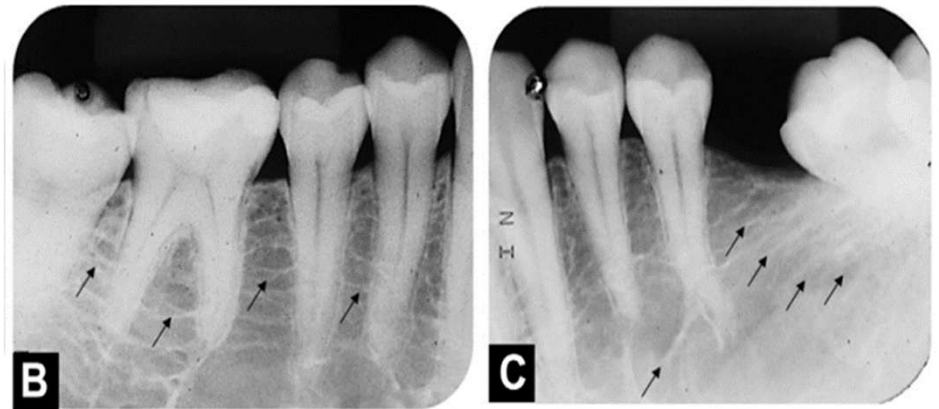
Sickle Cell Anemia

- Congenital disorder associated with structural changes in the hemoglobin protein.
- African Americans are the most affected ethnic group
- Physical Manifestation often due to blockage of small blood vessels
 - Arthritis, abdominal and chest pain
 - Swelling of hands and feet
 - Pallor or jaundice appearance
 - History of frequent infections and skin ulcers



Sickle Cell Anemia

- Oral Manifestations
 - Widening of the bony trabeculae
 - “Step-ladder appearance”
 - Delayed dental eruption
 - Dental hypoplasia



Oral Surg Oral Med Oral Pathol Oral Radiol 2017;124:76-84)

Sickle Cell Anemia



“Hair on end” radiographic appearance

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Glucose-6 Phosphatase Dehydrogenase Anemia (G6PD)

- Inherited defects of the enzymes needed to convert carbohydrates into energy in red blood cells.
- World's most common enzyme disorder
- Most affected patients are of Mediterranean, Middle Eastern, and Asian descent
- Physical Manifestations
 - Jaundiced skin, yellow discoloration of the sclera, splenomegaly
- Oral Manifestations
 - Pallor of the oral mucosa

Aplastic Anemia

- Anemia caused by depletion of all types of hematopoietic cells due to bone marrow suppression
- Physical manifestations
 - Pallor, fatigue, history of infection,
 - Spontaneous bleeding
- Oral manifestations
 - Gingival hyperplasia, spontaneous hemorrhage (petechiae, purpura, and ecchymosis), ulceration, periodontal disease and pallor of the oral soft tissues



Neville BW, Damm DD, Allen CM et al. Oral and maxillofacial pathology, ed 3, St. Louis. 2009 Saunders

Cobalamin (B12) and Folate (B9) Deficiency

- Causes
 - Nutritional deficiencies
 - Malabsorption
 - Alcohol and drug abuse
 - Therapeutic Drugs
- **Pernicious anemia**
 - Vitamin B12 (cobalamin) deficiency
 - Autoimmune condition associated with the atrophy of gastric mucosa

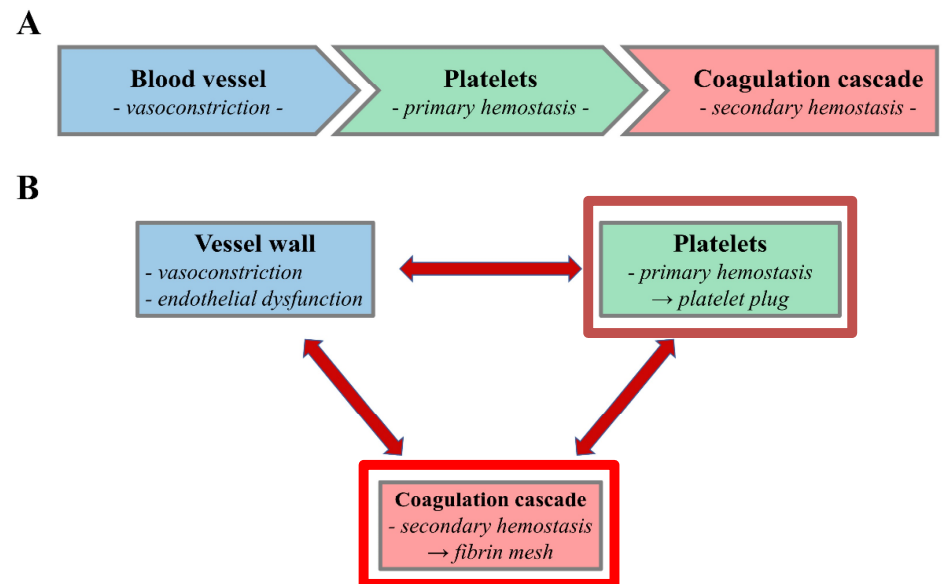
Cobalamin (B12) and Folate (B9) Deficiency

- Physical Manifestations
 - Pallor, fatigue, weakness, abnormal gait, ataxia, tingling and numbness in extremities (**paresthesia**), cognitive impairment and psychiatric changes
- Oral Manifestations
 - Atrophic glossitis, glossydynia, angular cheilitis, aphthous-like ulceration

Bleeding Disorders

There are a wide variety of bleeding disorders that can affect patient care in the dental practice.

- Hemostasis
 - Vasoconstriction
 - Platelet plug formation
 - Coagulation
 - Fibrinolysis



***Table 25.2 Congenital Bleeding Disorders**

Platelet Bleeding Disorders

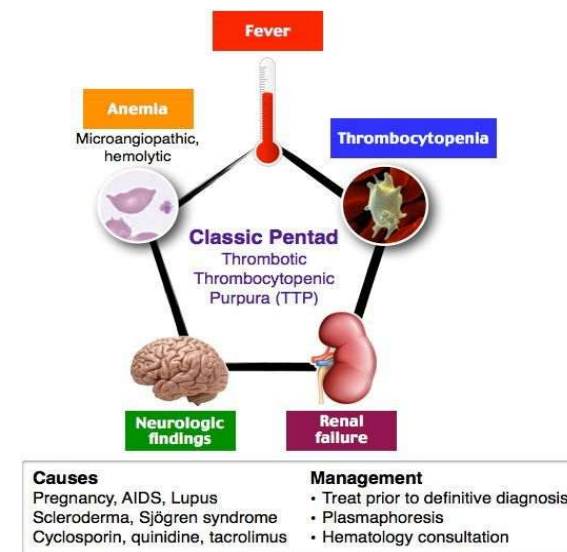
- Thrombocytopenia
 - Thrombotic Thrombocytopenic Purpura (TTP)
 - Idiopathic Thrombocytopenic Purpura (ITP)
- Drug-Induced Platelet Disorders
- Von Willebrand Disease (vWD)



www.slideshare.net/VijayShankar4/bleeding-disorders-62345706

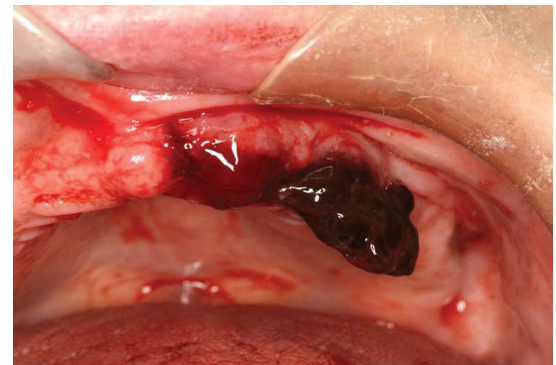
Thrombocytopenia

- Thrombotic Thrombocytopenic Purpura (TTP)
 - Fever, **thrombocytopenia**, hemolytic anemia, renal failure, and neurologic findings
 - Low ADAMTS13 in the body
- Idiopathic Thrombocytopenic Purpura (ITP)
 - Development of antibodies against platelets

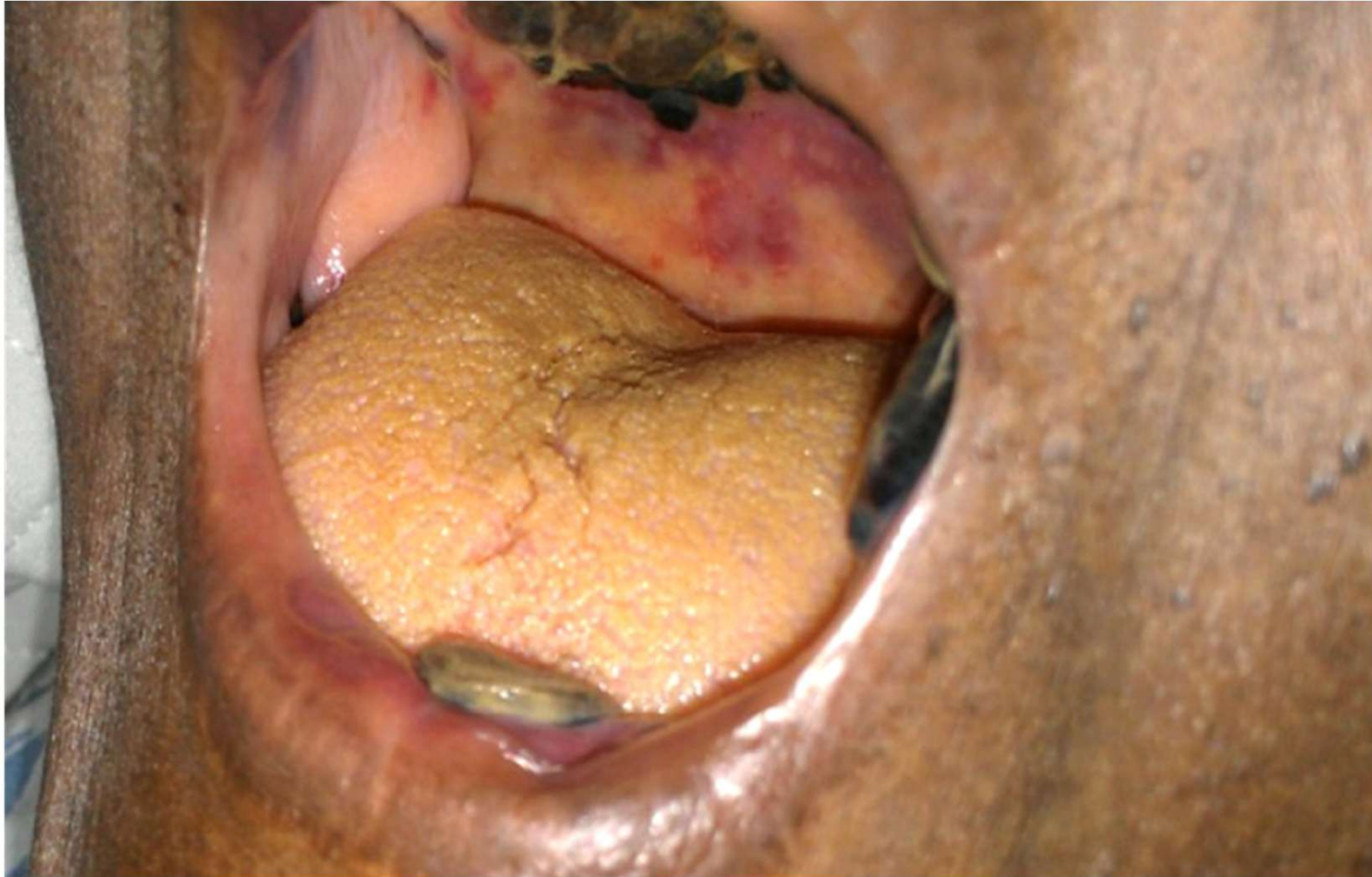


Physical and Oral Manifestations

- Physical
 - Pallor, jaundiced appearance, a history of epistaxis, and areas of bruising noted on the skin
- Oral
 - Spontaneous bleeding of the gingiva
 - Submucosal hemorrhage in the form of petechiae, purpura, ecchymosis, and hematomas
 - Hemarthrosis
 - Liver clots
 - Excessive bleeding after dental procedures



Idiopathic Thrombocytopenia Purpura



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Drug-Induced Platelet Disorders

(Antiplatelets agents Table 24.2)

- Drugs with antiplatelet activity
 - Aspirin
 - Nonsteroidal anti-inflammatory drugs (NSAIDs)
 - Presantine[®] } Dipyridamole
 - Aggrenox[®] }
 - Platelet adenosine diphosphate (ADP) receptor agonist
 - Plavix[®] (Clopidogrel)
 - Effient[®] (Prasurgel)
 - Brilinta[®] (Ticagrelor)
 - Ticlid[®] (Toclopidine)

von Willebrand Disease (vWD)

- Autosomal disorder caused by a deficient or defective plasma von Willebrand factor (vWF)
- Three types
 - Type 1
 - Decreased production of vWF
 - Type 2
 - Defective vWF
 - Type 3
 - Absence of vWF

von Willebrand Disease (vWD)

- Physical and Oral Manifestations
 - Mild forms-bleeding after surgery
 - Patients with this form may have a negative history for bleeding issues.
 - Severe forms-Epistaxis, spontaneous mucosal bleeding, and hemarthroses



Journal of Contemporary Dentistry, May-August 2015;5(2):107-112

Coagulation Disorders

- Hemophilia A
- Hemophilia B
- Disseminated Intravascular Coagulation (DIC)
- Liver Disease
- Drug-Induced Coagulation Disorders

Hemophilia

- Sex-linked coagulation disorder with a male predilection
- Hemophilia A
 - Deficiency of factor VIII in the coagulation cascade
- Hemophilia B
 - Deficiency of factor IX
- Oral Manifestations
 - Spontaneous bleeding and
 - Hemarthrosis



Disseminated Intravascular Coagulation (DIC)

Uncontrolled thrombus formation and bleeding

Causes: **(MOST)**

M- Malignancy/Misc.

O- Obstetric Complications

S- Sepsis (severe infection)/Severe autoimmune disorders

T- Trauma

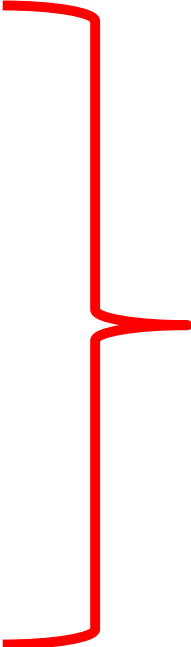
Liver Disease

- Severe liver diseases can increase the risk of hemorrhage
 - Many clotting proteins are synthesized in the liver
 - Factors II, VII, IX, and X
- In advanced liver disease there is:
 - Decreased clearance of activated clotting factors
 - Decreased platelet production
 - Increased platelet destruction by the spleen

Drug-Induced Coagulation Disorders

(Anticoagulants Table 24.2)

- Warfarin (Coumadin[®])
- Heparins
- Factor Xa inhibitor
 - Fondaparinux (Arixtra[®])
 - Rivaroxaban(Xarelto[®])
 - Apixaban (Eliquis[®])
- Factor IIa inhibitor
 - Dabigatran (Pradaxa[®])



No Lab test to
assess the ability
to clot.

Clinical Considerations for patients with medical disorders

- Type and severity of disorder
- Date of diagnosis
- Signs and symptoms are associated with the disorder
- Is the patient seeing their physicians/specialist on a regular basis?
- Treatment
 - Current or past treatment for the disorder
 - Patient compliance
 - Is the disease well or poorly controlled?

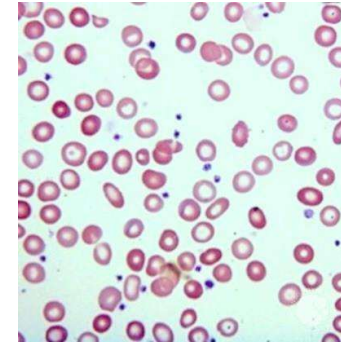
Dental Treatment Consideration

- Is the patient safe to treat in an outpatient setting?
- Routine *versus* emergency dental treatment
- What complications can occur during or after dental treatment?
 - Is there an increased risk of bleeding or infection associated with dental treatment?
 - Are there any risks associated with anesthesia (local and/or general), sedation or nitrous oxide usage ?
 - Is the patient able tolerate over-the-counter or prescription medications?

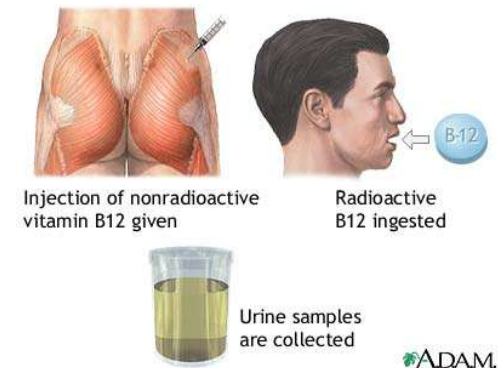
Laboratory Testing

* Blood test for anemia – Table 22.4 textbook

- CBC w/ Differential
- Specific test for vitamin/mineral/antibody levels
- INR
- Peripheral Blood Smear
- Bone marrow aspirate
- Bilirubin
- Schilling's test
- Erythropoietin levels



Peripheral Blood Smear



Complete Blood Count with Differential

Table 8.5. Normal Complete Blood Cell (CBC) Count

Blood Cell Type	Normal Reference Range
Red blood cells ^b (RBCs)	M: 4.3–5.7 million cells/ μ L F: 3.8–5.1 million cells/ μ L
Hemoglobin ^b (Hgb)	M: 13.5–17.5 g/dL F: 12.0–16.0 g/dL
Hematocrit ^b (HCT)	M: 39–49% F: 35–45%
Platelets	150,000–400,000/ mm^3
White blood cells (WBCs) ^c	4,500–11,000 cells/ μ L

Differential WBC

Neutrophils segmented (PMNs)	54–62%	3,000–5,800/ mm^3
Neutrophils bands	3–5%	150–400/ mm^3
Lymphocytes	23–33%	1,200–3,000/ mm^3
Monocytes	3–7%	285–500/ mm^3
Eosinophils	1–3%	50–250/ mm^3
Basophils	0–0.75%	15–50/ mm^3

Dental Modifications for Anemias

- Poorly controlled iron deficiency anemia
 - Review laboratory results and consult with patient's medical doctor
- Hemolytic Anemias
 - Be mindful of medications that can affect oxygen saturation
 - Is antibiotic prophylaxis required?
 - Predisposition to infection
 - Local anesthesia considerations
 - Short dental appointments to minimize stress
 - Medications.

Dental Modifications of Anemias

- Aplastic Anemia
 - Medical consultation
 - Platelet infusion
- Vitamin B12 and Folate Deficiencies (Megablastic Anemia)
 - Poorly controlled disease
 - Measures put in place to help patients with essential tremors

Dental Modifications for Bleeding Disorders

- INR laboratory values
 - Coumadin
- Discontinuation of antiplatelet/anticoagulant therapy for **major** procedures
- Clotting concentrates for hemophilic patients (Table 25.3)
- Local measure to help with hemostasis after treatment (Table 25.6)
 - Suture
 - Absorbable collagen products
 - Topical chemical agents

Laboratory Values

Table 9.3. Common Laboratory Tests Used to Assess Hemostasis

Laboratory Test	Normal Range	What It Measures
Platelet count	150,00–400,00 cells/mL	Platelet quantity
Ivy bleeding time (BT)	<6 minutes	Platelet function (quantity and quality)
PFA-100	Closure time <193 seconds	Quantitative and qualitative measurement of platelet adhesion, activation, and aggregation
Prothrombin time (PT)	11–14 seconds	Factors II (prothrombin), V, VII, and X, and fibrinogen
International normalized ratio (INR)	1.0	
Activated partial thromboplastin time (aPTT)	27–38 seconds	Factors II, V, VIII, IX, X, XI, and XII
Thrombin time (TT)	9–13 seconds	Abnormalities in the conversion of fibrinogen to fibrin

PFA-100, platelet function analyzer 100.

The ADA Practical Guide to Patients with Medical Conditions

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Dental Procedure	Suboptimal INR Range		Normal Target INR Range			Out of Range
	<1.5	1.5 to <2.0	2.0 to <2.5	2.0 to 3.0	> 3.0 to 3.5	> 3.5
				Mechanical Prosthetic Heart Valves		
			Atrial Fibrillation; Venous Thrombosis; Pulmonary or Systemic Embolism; Acute MI			
Examination, radiographs, impressions, orthodontics						
Simple restorative dentistry, supragingival prophylaxis						
Complex restorative dentistry, scaling & root planing, endodontics					Probably safe	
Simple extraction, curettage, gingivoplasty, biopsy				Local measures§	Local measures§	
Multiple extractions, single bony impaction extraction			Local measures§	Local measures§	Local measures§	
Gingivectomy, minor periodontal flap surgery,apicoectomy, single implant placement	Probably safe	Probably safe	Probably safe			
Full mouth or full arch extractions	Probably safe	Local measures§				
Extensive flap surgery, extraction of multiple bony impactions, multiple implant placement	Probably safe					

INR= International Normalized Ratio; MI= Myocardial Infarction. Green indicates that it is safe to proceed in a routine manner (local factors such as periodontitis/gingival inflammation can increase severity of bleeding; the clinician should consider all factors when making a risk assessment). Yellow, use caution, but in many instances the procedure can be safely performed with judicious use of local measures. Red, procedures not advised at current INR level; refer to physician for Coumadin® adjustment.

§ Increased need for use of local measures such as sutures, oxidized cellulose, microfibrillar collagen hemostat, topical thrombin and/or epsilon aminocaproic acid or tranexamic acid. (Herman WW, Konzelman Jr JL, Sutley SH. Current perspectives on dental patients receiving coumarin anticoagulant therapy. JADA 1997;128(3):327-35. Copyright © 1997 American Dental Association. All rights reserved. Adapted 2011 with permission of the American Dental Association.)

Figure 9.4 Safety of outpatient dental procedures for patients on Coumadin®.

The ADA Practical Guide to Patients with Medical Conditions
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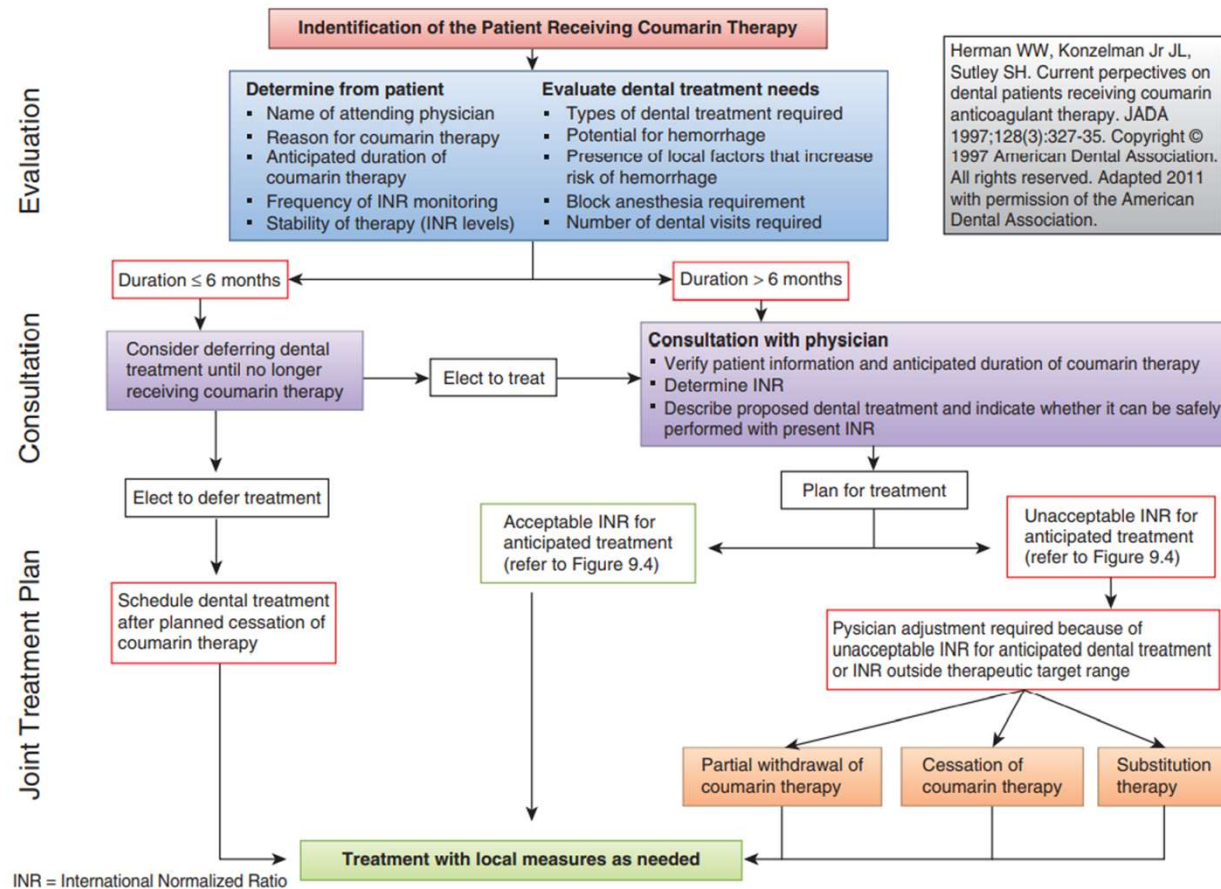


Figure 9.5 Algorithm for nonurgent dental care of patients receiving coumarin therapy.

Dental Modifications for Bleeding Disorders

- Medical treatments for bleeding disorders
 - Transfusions and coagulation factor replacement therapy
 - Desmopressin
 - Prednisone
 - Intravenous gammaglobulin (IVIG)
- Liver function
- Drug interactions

Resources

